Power Series and Taylor Series

Radius and interval of convergence

1 Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{n}$.

2 Find the radius of convergence of $\sum_{n=0}^{\infty} \frac{x^n}{n!}$.

Taylor and Maclaurin series

3 Find the first three nonzero terms of the Maclaurin series for $f(x) = e^x$, using $f^{(n)}(0) = 1$ for all n.

4 Find the third-degree Taylor polynomial for $f(x) = \cos x$ centered at $x = 0$.

5 Use the Maclaurin series $\sin x = x - \frac{x^3}{6} + \frac{x^5}{120} - \cdots$ to approximate $\sin(0.2)$ using the first two nonzero terms, to four decimal places.

Interval of convergence (including endpoints)

6 Find the interval of convergence of $\sum_{n=1}^{\infty} \frac{(x-2)^n}{n}$, checking both endpoints.

7 Find the radius of convergence of $\sum_{n=1}^{\infty} n!\, x^n$.

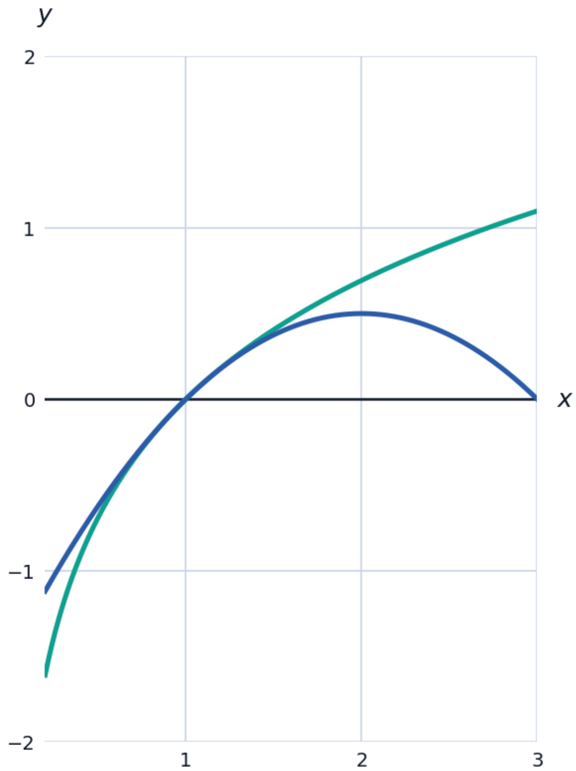
Using known series

8 Using the known Maclaurin series $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$, write the Maclaurin series for e^{-x^2} (substitute $-x^2$ for x), giving the first three terms.

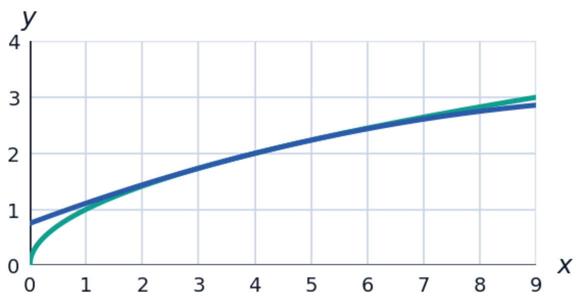
9 Using $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$ for $|x| < 1$, write the power series for $\frac{1}{1+x^2}$ (substitute $-x^2$ for x).

Taylor series centered at a nonzero point

10 Find the first three terms of the Taylor series for $f(x) = \ln x$ centered at $x = 1$, using $f(1) = 0$, $f'(1) = 1$, $f''(1) = -1$.



11 Find the second-degree Taylor polynomial for $f(x) = \sqrt{x}$ centered at $x = 4$, given $f(4) = 2$, $f'(4) = \frac{1}{4}$, $f''(4) = -\frac{1}{32}$.

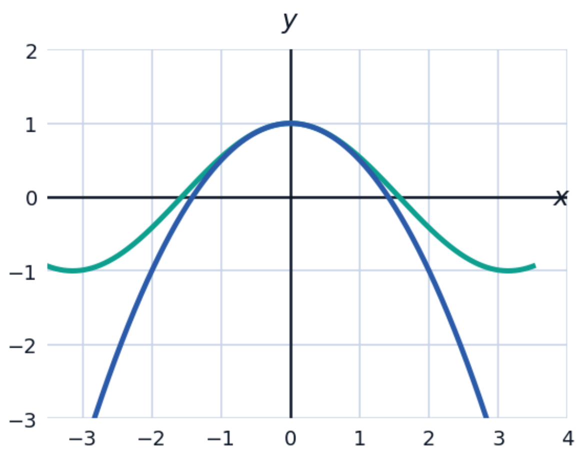


Error estimation with Taylor polynomials

12 Using the second-degree Maclaurin polynomial for $\cos x$, $T_2(x) = 1 - \frac{x^2}{2}$, estimate $\cos(0.3)$ to four decimal places, and compare to the true value $\cos(0.3) \approx 0.9553$.

Visualizing a Taylor approximation

13 The graph compares $f(x) = \cos x$ to its second-degree Maclaurin polynomial $T_2(x) = 1 - \frac{x^2}{2}$. Use the graph to describe where the approximation is best and where it breaks down.



Differentiating and integrating power series

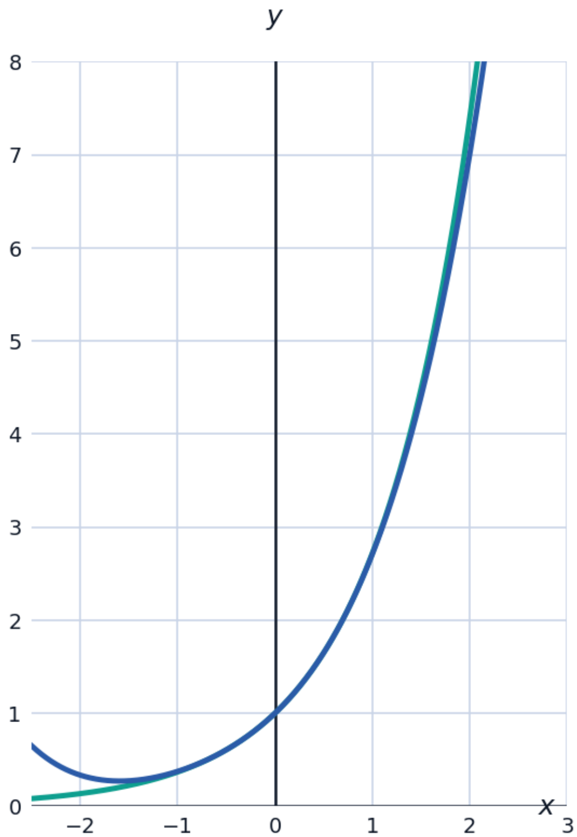
14 Given $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$ for $|x| < 1$, differentiate both sides term-by-term to find a power series for $\frac{1}{(1-x)^2}$.

15 Using the Maclaurin series $\sin x = \sum^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$, integrate term-by-term to find the first three terms of the power series for $\int \sin x \, dx = -\cos x + C$, and confirm they match the known Maclaurin series for $-\cos x$.

Using Taylor series to evaluate a limit

16 Use the Maclaurin series $\sin x = x - \frac{x^3}{6} + \cdots$ to evaluate $\lim_{x \to 0} \frac{\sin x - x}{x^3}$.

17 Find the fourth-degree Taylor polynomial for $f(x) = e^x$ centered at $x = 0$, and use it to estimate $e^{0.5}$, comparing to the true value $e^{0.5} \approx 1.6487$.



18 An engineer needs a quick approximation for $\ln(1.1)$ without a calculator. Using the Maclaurin series $\ln(1 + x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots$ for $|x| < 1$, find the approximation using the first three terms with $x = 0.1$, and compare to the true value $\ln(1.1) \approx 0.09531$.